

JOONHYUNG KWAK

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SUMMARY

M.S. candidate in Artificial Intelligence at Sungkyunkwan University, graduating Aug. 2026. Specializing in **LLM inference optimization and causal reasoning systems for mental-health applications**. Demonstrated ability to take research prototypes to production: scaled vLLM inference pipelines on 8K+ user corpora, designed statistically rigorous LLM evaluation, and built end-to-end crawling-to-insight systems consumed by industry stakeholders.

EDUCATION

M.S. in Artificial Intelligence Sungkyunkwan University Advisor: Prof. JinYeong Bak (Human-Language Intelligence Lab) Thesis: <i>From Post to Temporal Causality: Constructing Temporal Causal Symptom Graphs Using Large Language Models</i>	Mar. 2024 – Aug. 2026 (Expected)
Exchange Student in Computer Science Deggendorf Institute of Technology, Germany Advisor: Prof. Patrick Glauner	Mar. 2022 – Jul. 2022
B.S. in Information and Communication Engineering Inha University Exchange Semester: Deggendorf Institute of Technology, Germany (Mar. – Jul. 2022)	Mar. 2020 – Feb. 2024

TECHNICAL SKILLS

LLM Serving	vLLM (PagedAttention, continuous batching), HuggingFace Transformers, LoRA/PEFT, prompt engineering, LLM-as-a-Judge evaluation
Causal ML	Bootstrapped ATE estimation, confidence interval analysis, temporal causal graph construction
Deep Learning	PyTorch, PyTorch Lightning (DDP multi-GPU), TensorFlow
Data Engineering	Playwright (async crawling, anti-bot handling), asyncio, pandas, MySQL, AWS
Backend	REST/WebSocket APIs, multi-agent LLM orchestration, Docker, Git
Languages	Python (advanced), C/C++, Korean (native), English (fluent)

RESEARCH EXPERIENCE

Human-Language Intelligence Lab, Sungkyunkwan University <i>Graduate Researcher (Advisor: Prof. JinYeong Bak)</i>	Mar. 2024 – Present
- Constructed temporal causal symptom graphs using LLM from 230K+ SNS posts by designing a day-level clustering preprocessing pipeline, replacing a legacy causal-matrix approach that failed on sparse user timelines (80% coverage improvement). - Scaled LLM inference throughput by 6× on an 8K-user corpus using vLLM with PagedAttention and continuous batching, reducing end-to-end labeling time from 9h to 1.5h on 2×A6000. - Designed an LLM-as-a-Judge evaluation framework with bootstrapped ATE and 95% confidence intervals (10k resamples), enabling statistically grounded comparison of causal graph variants rather than point estimates. - Trained multi-label symptom classifiers with PyTorch Lightning DDP on 2 GPUs, achieving 0.78 macro-F1 on PsySym dataset — a 8-point gain over few-shot GPT-4o baselines.	

- (Sub-project: *Mental Health Screening from Military App Logs, Jun. 2024 – Jul. 2025*) Processed longitudinal diary-app logs from 214 active-duty soldiers over 2 months; built an encoder-based sequence model that outperformed few-shot Solar by 9 F1 points, demonstrating structured behavioral signals beat prompt-only LLMs on low-resource clinical tasks.

Biomedical Signal Lab, SNU Bundang Hospital

Sep. 2024 – Oct. 2024

Research Intern

- Built an ECG/ABP preprocessing pipeline (Pan-Tompkins QRS detection, cubic-spline baseline-wander removal via R-peak-anchored knots, resampling to 100 Hz with per-beat normalization to 128 samples) on 1,500 patient records, with quality-filtering rules rejecting 30% of noisy segments.
- Developed a multi-input 1D Inception-CNN in PyTorch fusing waveform features with patient demographics via late concatenation for stroke risk prediction, reporting ROC-AUC of 0.81.
- Fine-tuned Phi-2 (2.7B) on the Asclepius clinical-notes dataset via 4-bit NF4 QLoRA for clinical question answering, deploying a Gradio-based A/B comparison demo against the vanilla Phi-2 baseline.

Data Intelligence Lab, Inha University

Nov. 2022 – Dec. 2023

Undergraduate Researcher

- Improved Korean ASR word-error rate from 12.2% to 8.3% by building a text-normalization pipeline handling numerals, loanwords, and disfluencies before Transformer-based decoding.
- Implemented custom Transformer variants in PyTorch for solar-power forecasting, improving MAPE by 7% over the vanilla baseline.

Nursing Informatics Lab, Inha University

Aug. 2022 – May 2023

Research Assistant

- Extracted clinical keyword trends from 5K EMR notes using N-gram tokenization and word2vec clustering.
- Fine-tuned a domain-specific BERT for medical NER, achieving F1 of 82.8 on 23 entity types.

SELECTED PROJECTS

Xueqiu Investor Sentiment Pipeline

Nov. 2025 – Dec. 2025

Independent Project — Hedge Fund Data Engineering Assignment

- Built an end-to-end async crawler (Playwright + asyncio) handling dynamic rendering, infinite scroll, and anti-bot detection across 7 market tabs on Xueqiu, collecting 700 posts.
- Integrated Fireworks AI (Llama-3) for CN→EN translation and sentiment classification.
- Delivered structured Markdown analyst reports consumed by hedge-fund investment team with full project open-sourced on GitHub with English documentation.

Suicide & Self-harm Content Detection on SNS

Sep. 2024 – May 2025

Graduate Research (Clinical Collaboration)

- Designed IRB-compliant data collection protocol; recruited 128 participants from clinical settings and collected 3k real SNS posts with text + image metadata.
- Built a GPT-4o structured-output pipeline with hierarchical category taxonomy (6 categories).

Real-time Multi-Agent LLM Debate System

Mar. 2024

ITRC Project

- Architected a real-time multi-agent orchestration backend streaming LLM-to-LLM debate turns over Web-Socket, coordinating OpenAI and Gemini agents with role-conditioned prompts and turn-taking state management.
- Handled concurrent token-level streaming and solved agent interleaving deadlocks by introducing a priority queue arbitrator.

Audio Emotion Classification with Dynamic Attention Windows

Feb. 2023 – Jun. 2023

Undergraduate Thesis — Published at KCC 2023

- Identified that fixed-window audio preprocessing discards 25% of emotion-bearing speech segments by visualizing internal attention maps.
- Designed an attention-driven dynamic windowing method that improved classification accuracy from 78% to 84% on AIHub emotion speech corpus.

DeinHaus 4.0 — Smart-Home Anomaly Detection

Feb. 2022 – Aug. 2022

Industry-linked Project, Deggendorf Institute of Technology

- Analyzed multi-sensor ventilation and activity streams from 5 instrumented aging-in-place households to build automated control.

TEACHING

Teaching Assistant, *Introduction to Artificial Intelligence*, SKKU (Autumn 2024)

Teaching Assistant, National Defense AI Education Program (Summer 2024)

Teaching Assistant, AI Special Lecture for Non-major Faculty, SKKU (Summer 2024, Winter 2025)

AWARDS

Research Scholarship for Undergraduate Researchers, Inha University (2023)

Han-eum ICT Contest Award (2021)

CONFERENCES

J. Kwak et al., “Dataset refining techniques for improving training efficiency of voice-based situation classification models,” Korea Computer Congress (KCC), Poster, 2023.